

► narrow set of applications for his theory. Most of these settings were military in nature. One such was the idea of mutually assured destruction, in which equilibrium is reached by arming adversaries with nuclear weapons (some have suggested that the film character of Dr Strangelove was based on von Neumann). None of this was particularly useful for thinking about situations—including most types of market—in which one party's victory does not automatically imply the other's defeat.

Even so, the economics profession initially shared von Neumann's assessment, and largely overlooked Nash's discovery. He threw himself into other mathematical pursuits, but his huge promise was undermined when in 1959 he started suffering from delusions and paranoia. His wife had him hospitalised; upon his release, he became a familiar figure around the Princeton campus, talking to himself and scribbling on blackboards. As he struggled with ill health, however, his equilibrium became more and more central to the discipline. The share of economics papers citing the Nash equilibrium has risen sevenfold since 1980, and the concept has been used to solve a host of real-world policy problems.

One famous example was the American hospital system, which in the 1940s was in a bad Nash equilibrium. Each individual hospital wanted to snag the brightest medical students. With such students particularly scarce because of the war, hospitals were forced into a race whereby they sent out offers to promising candidates earlier and earlier. What was best for the individual hospital was terrible for the collective: hospitals had to hire before students had passed all of their exams. Students hated it, too, as they had no chance to consider competing offers.

Despite letters and resolutions from all manner of medical associations, as well as the students themselves, the problem was only properly solved after decades of tweaks, and ultimately a 1990s design by Elliott Peranson and Alvin Roth (who later won a Nobel economics prize of his own). Today, students submit their preferences and are assigned to hospitals based on an algorithm that ensures no student can change their stated preferences and be sent to a more desirable hospital that would also be happy to take them, and no hospital can go outside the system and nab a better employee. The system harnesses the Nash equilibrium to be self-enforcing: everyone is doing the best they can based on what everyone else is doing.

Other policy applications include the British government's auction of 3G mobile-telecoms operating licences in 2000. It called in game theorists to help design the auction using some of the insights of the

The prisoner's dilemma

Prisoner A		Prisoner B	
Confess	Confess	Both go to jail for ten years	Prisoner B gets life imprisonment, A goes free
	Keep quiet	Prisoner A gets life imprisonment, B goes free	Both go to jail for one year
Keep quiet	Confess	Both go to jail for one year	Prisoner B gets life imprisonment, A goes free
	Keep quiet	Both go to jail for ten years	Prisoner B gets life imprisonment, A goes free

Nash equilibrium, and ended up raising a cool £22.5 billion (\$35.4 billion)—though some of the bidders' shareholders were less pleased with the outcome. Nash's insights also help to explain why adding a road to a transport network can make journey times longer on average. Self-interested drivers opting for the quickest route do not take into account their effect of lengthening others' journey times, and so can gum up a new shortcut. A study published in 2008 found seven road links in London and 12 in New York where closure could boost traffic flows.

Game on

The Nash equilibrium would not have attained its current status without some refinements on the original idea. First, in plenty of situations, there is more than one possible Nash equilibrium. Drivers choose which side of the road to drive on

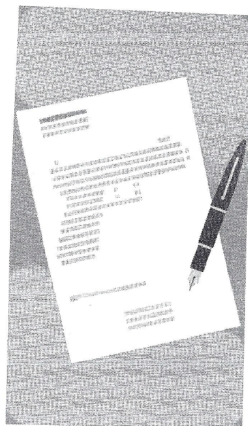
as a best response to the behaviour of other drivers—with very different outcomes, depending on where they live: they stick to the left-hand side of the road in Britain, but to the right in America. Much to the disappointment of algebra-toting economists, understanding strategy requires knowledge of social norms and habits. Nash's theorem alone was not enough.

A second refinement involved accounting properly for non-credible threats. If a teenager threatens to run away from home if his mother separates him from his mobile phone, then there is a Nash equilibrium where she gives him the phone to retain peace of mind. But Reinhard Selten, a German economist who shared the 1994 Nobel prize with Nash and John Harsanyi, argued that this is not a plausible outcome. The mother should know that her child's threat is empty—no matter how tragic the loss of a phone would be, a night out on the streets would be worse. She should just confiscate the phone, forcing her son to focus on his homework.

Mr Selten's work let economists whittle down the number of possible Nash equilibria. Harsanyi addressed the fact that in many real-life games, people are unsure of what their opponent wants. Economists would struggle to analyse the best strategies for two lovebirds trying to pick a mutually acceptable location for a date with no idea of what the other prefers. By embedding each person's beliefs into the game (for example that they correctly think the other likes pizza just as much as sushi), Harsanyi made the problem solvable. A different problem continued to lurk. The predictive power of the Nash equilibrium relies on rational behaviour. Yet humans often fall short of this ideal. In experiments replicating the set-up of the prisoner's dilemma, only around half of people chose to confess. For the economists who had been busy embedding rationality (and Nash) into their models, this was problematic. What is the use of setting up good incentives, if people do not follow their own best interests?

All was not lost. The experiments also showed that experience made players wiser; by the tenth round only around 10% of players were refusing to confess. That taught economists to be more cautious about applying Nash's equilibrium. With complicated games, or ones where they do not have a chance to learn from mistakes, his insights may not work as well.

The Nash equilibrium nonetheless boasts a central role in modern microeconomics. Nash died in a car crash in 2015; by then his mental health had recovered, he had resumed teaching at Princeton and he had received that joint Nobel—in recognition that the interactions of the group contributed more than any individual. ■



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